SIMATS ENGINEERING

SIMATS Online Programming Lab

C PROGRAMMING

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QUESTIONS 25 / 25 completed

Diagonal or Not

DIAGONAL OR NOT

Write a C program to check if a matrix is diagonal or not.

PROGRAM EDITOR

C | Run | Save

```c
#include <stdio.h>

int main() {
    int n, i, j, flag = 1;
    int matrix[10][10];  // supports up to 10x10 matrix

    printf("Enter the size of the matrix (n x n): ");
    scanf("%d", &n);

    printf("Enter the elements of the matrix:\n");
    for (i = 0; i < n; i++) {
        for (j = 0; j < n; j++) {
            scanf("%d", &matrix[i][j]);
        }
    }

    // Check if all non-diagonal elements are 0
    for (i = 0; i < n; i++) {
        for (j = 0; j < n; j++) {
            if (i != j && matrix[i][j] != 0) {
                flag = 0;
```

OUTPUT

```
Enter the size of the matrix (n x n): Enter the elements of the matrix:

The matrix is NOT Diagonal
```

INPUT

```
3
0 0 5
0 7 0
0 0 9
```

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